Steamboat Hills: Advisor Briefing

Where the database stands, what NDEP data shows, and the chloride sampling-frequency plan explained plainly

2026-10-08

Why this meeting

The database now has enough real, repeated field data to say something concrete about how the system behaves, not just where things are located. This document lays out three things worth 15-20 minutes of your time: (1) what the database currently shows about the system, (2) what the new NDEP records add, and (3) the statistical plan for figuring out how often chloride needs to be sampled to track the system in real time – explained without assuming a stats background.

1. Current state of the Steamboat database

A few numbers to anchor the conversation: **1,600 water-chemistry samples**, **269 wells**, and **304,659 temperature-logger readings** are now in one relational database, spanning real field sampling (2026), NDEP compliance records (2024-2026), and historical literature data reaching back to 1950 (Sorey & Colvard 1992, Mariner & Janik 1995, Skalbeck 2001). The timeline below places the current reactivation event against that full history.

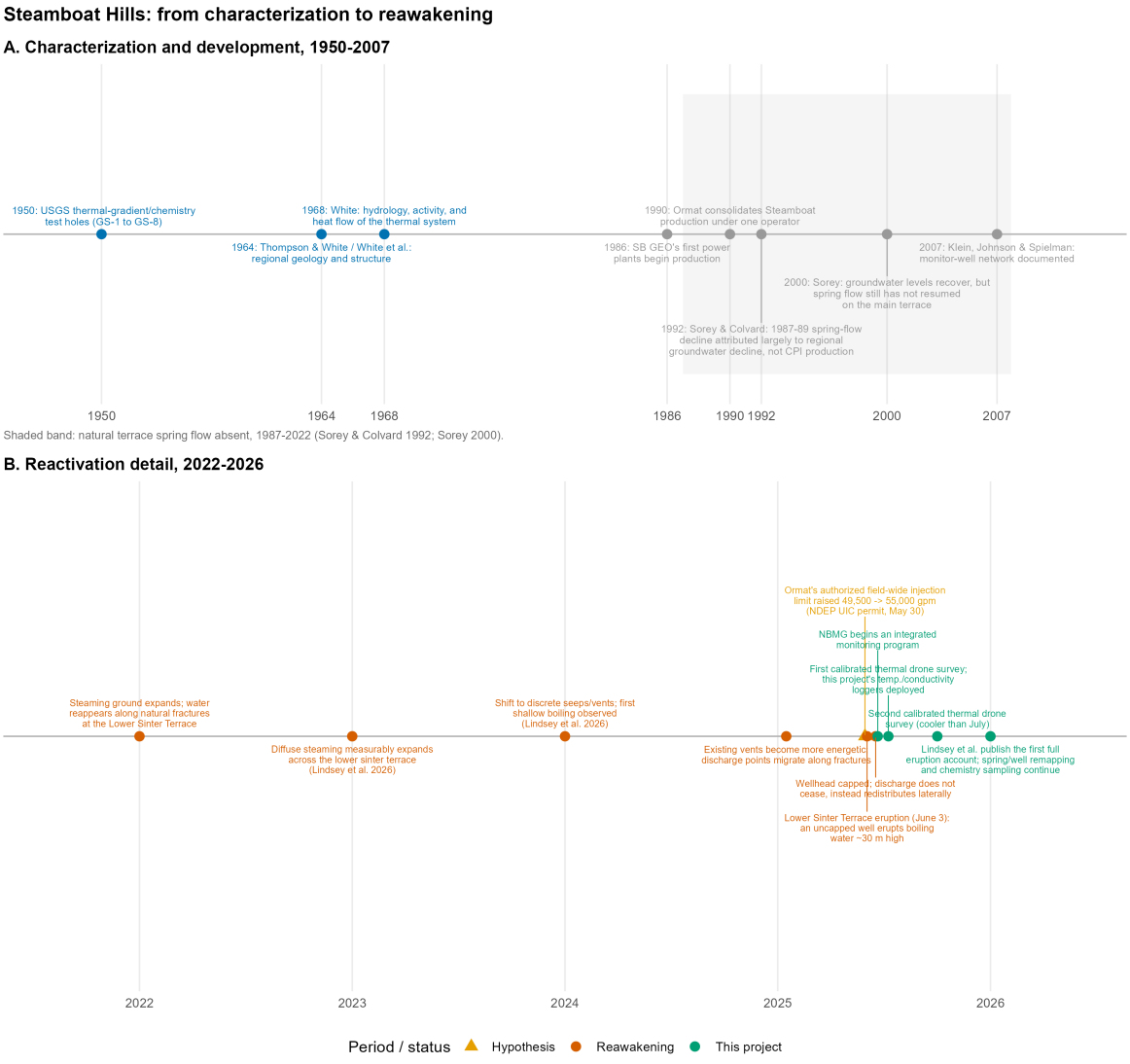


Figure 1: Seventy-year context: natural terrace-spring flow ceased around 1987, stayed absent through at least 2000 (Sorey 2000), and the 2022-2025 reactivation culminated in the June 2025 eruption. Ormat's permitted field-wide injection-limit increase (May 2025) is flagged as a candidate, unconfirmed contributing factor two weeks before the eruption – not a demonstrated cause.

Three real findings worth discussing directly:

- **A stable ~1.5-2 pH-unit gap between injected fluid and the shallow groundwater network**, holding across three real monthly sampling rounds (Sept-Dec 2025, 14

sites). This looks like water-rock buffering along a flow path, which argues for thinking about this system as reactive transport, not pure conservative mixing, at least for pH.

- **The Na/K geothermometer estimates a hotter reservoir (154-256 C) than the quartz/silica geothermometer (52-127 C)** for the same real thermal samples – consistent with a deep, equilibrated reservoir fluid that re-equilibrates silica faster than Na/K as it cools and dilutes en route to discharge.
- **Current chloride discharge to Steamboat Creek is roughly 27-59 L/s** (poster average ~27 L/s; one real dated measurement, 2026-05-01, gives 58.7 L/s), against a historical range of ~60-90 L/s for 2008-2016 and over 1,000 L/s reported in the 1950s-60s. The system is discharging far less than historically, even accounting for real seasonal variability.

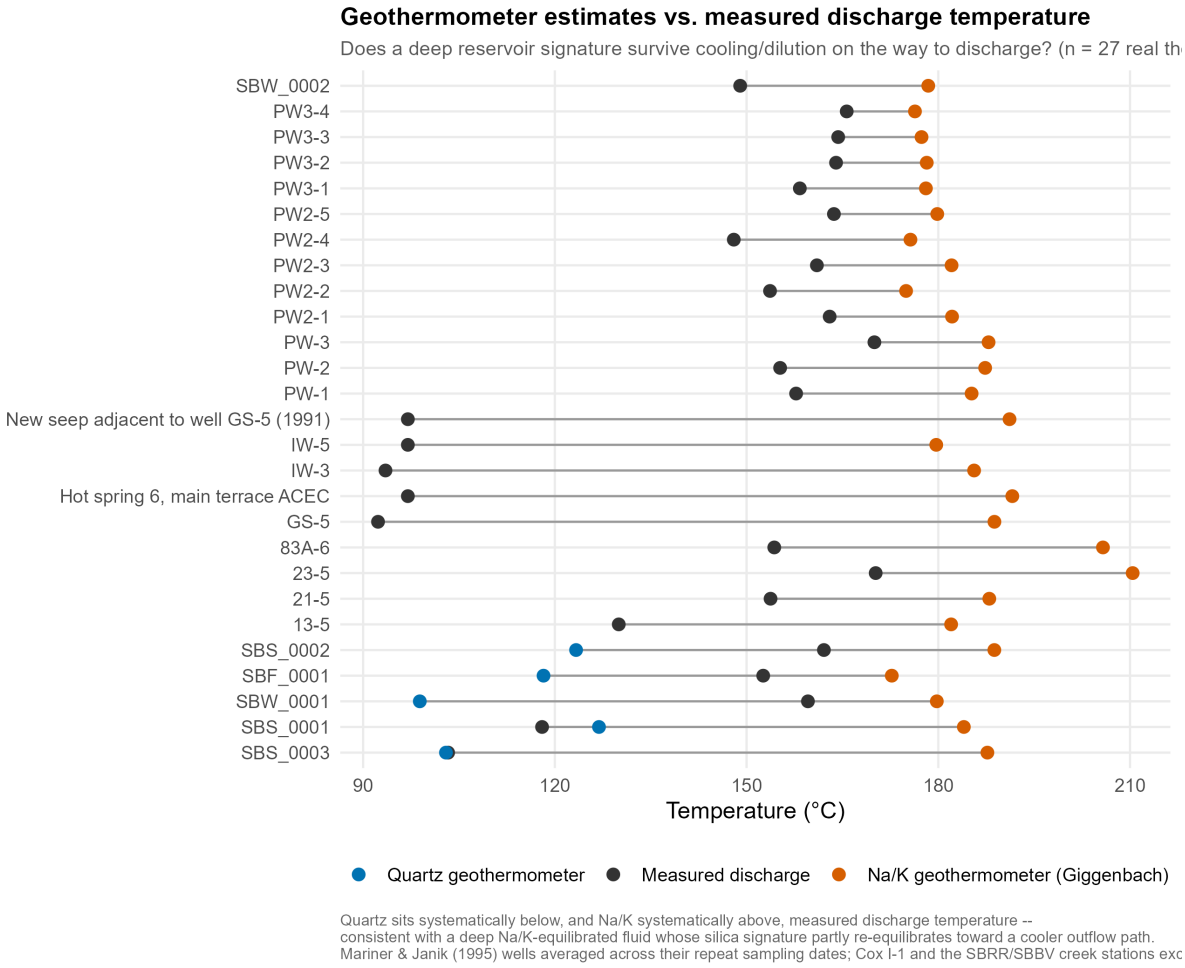


Figure 2: Na/K vs. quartz/chalcedony geothermometer divergence, averaged per well across 67 real geothermometer estimates.

2. What the NDEP records add

Scanned NDEP field-sampling forms (U230 forms + the 2024/2025/2026 TFT compliance reports) have been transcribed into the database over the last several weeks, turning one-off grab samples into a real, roughly monthly time series across **14 sites**: six named injection-side ports (Galena 1/2/3, SB2, SB3, SBHR) and eight shallow monitoring/domestic wells (Soccer Field, Herz Domestic, Herz Deep, Eich, NDOT, Boyd, Jeppson, Rogers).

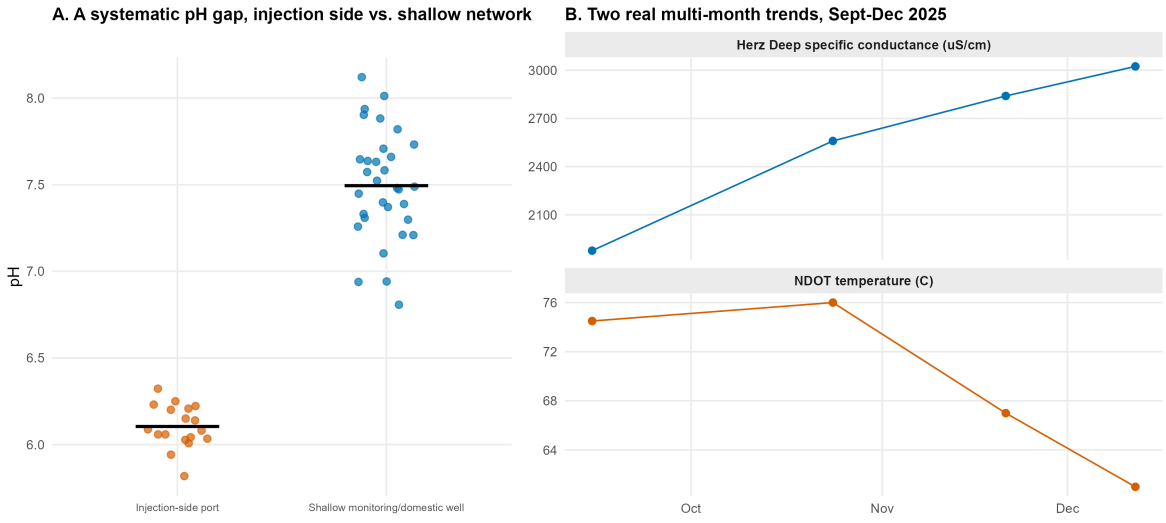


Figure 3: Left: pH, injection ports vs. shallow wells, pooled across three real rounds. Right: Herz Deep's conductance rising vs. NDOT's temperature falling over the same three months.

Two wells move in **opposite, uncorrelated directions** over the same three months: Herz Deep's conductance rises about 60% while its temperature stays flat; NDOT's temperature falls from ~76 C to ~61 C while its conductance stays flat. Neither temperature nor conductance alone is a safe universal stand-in for "thermal influence" at this site – worth a direct question to you about which parameter (or combination) is defensible to use that way.

NDEP's own UIC permit records also gave a real injection pressure/rate snapshot (one document, 2025-05-30) for all 12 named injection wells, including one well (23–33) whose reported current pressure already exceeds its own newly-authorized limit – flagged in the database, not resolved.

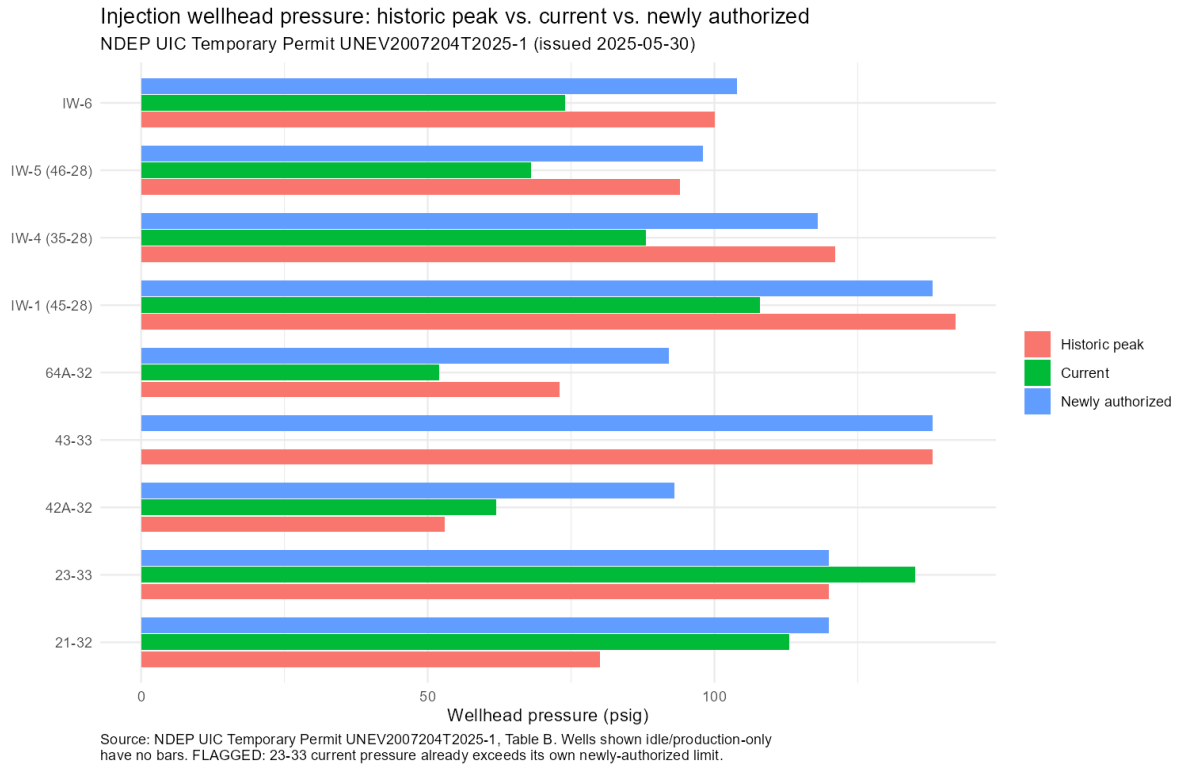


Figure 4: Historic-peak vs. current vs. newly-authorized wellhead pressure by injection well, per NDEP's May 2025 UIC permit.

3. The chloride sampling-frequency plan, explained plainly

The question: chloride concentration in Steamboat Creek is the best tracer of geothermal outflow, but lab chloride analysis is slow and expensive, so it can only be measured occasionally (a few times a month at most). Electrical conductivity, by contrast, can be measured continuously and cheaply with an in-stream sensor. **If conductivity tracks chloride closely enough, a continuous conductivity record can stand in for chloride between lab samples.** The open question is: how often does chloride actually need to be measured in the lab to keep that stand-in trustworthy?

The approach, in five plain steps (no statistics background needed):

1. **Start with a real, continuous conductivity record.** Two loggers have been running since July 2026 (one upstream/"control" at Rhodes Road, one downstream at Geiger Grade), recording every 15 minutes – a genuinely dense, real dataset.
2. **Build a model that predicts chloride from conductivity.** Once enough real chloride lab samples exist that overlap the logger record, a simple statistical model (basically:

"as conductivity goes up, chloride tends to go up too, by about this much") is fit to the paired data. This turns the continuous conductivity record into a continuous *estimated* chloride record.

3. **Pretend you only sampled chloride occasionally.** This is the core trick: take that estimated continuous chloride record and throw most of it away, keeping only points spaced a candidate interval apart – say, one chloride sample every day, or every 3 days, every week, every 2 weeks, every month.
4. **Reconstruct and compare.** Connect the dots between the kept points (linear interpolation) and compare that reconstructed curve back against the real, full continuous record. The gap between the two tells you how much error that sampling interval introduces.
5. **Repeat many times, randomizing the starting point (Monte Carlo).** A single test at "sample every 7 days" could get lucky or unlucky depending on *which* day of the week you happened to sample. So the test is repeated 200 times, each time starting on a different random day, and the errors are averaged. This produces a realistic, honest answer instead of one that depends on a lucky or unlucky draw.

The output is a simple curve: sampling interval on the x-axis, average reconstruction error on the y-axis. Wherever that curve crosses an acceptable error threshold (e.g. 5%) is the recommended maximum sampling interval – the coarsest schedule that still keeps the chloride estimate trustworthy.

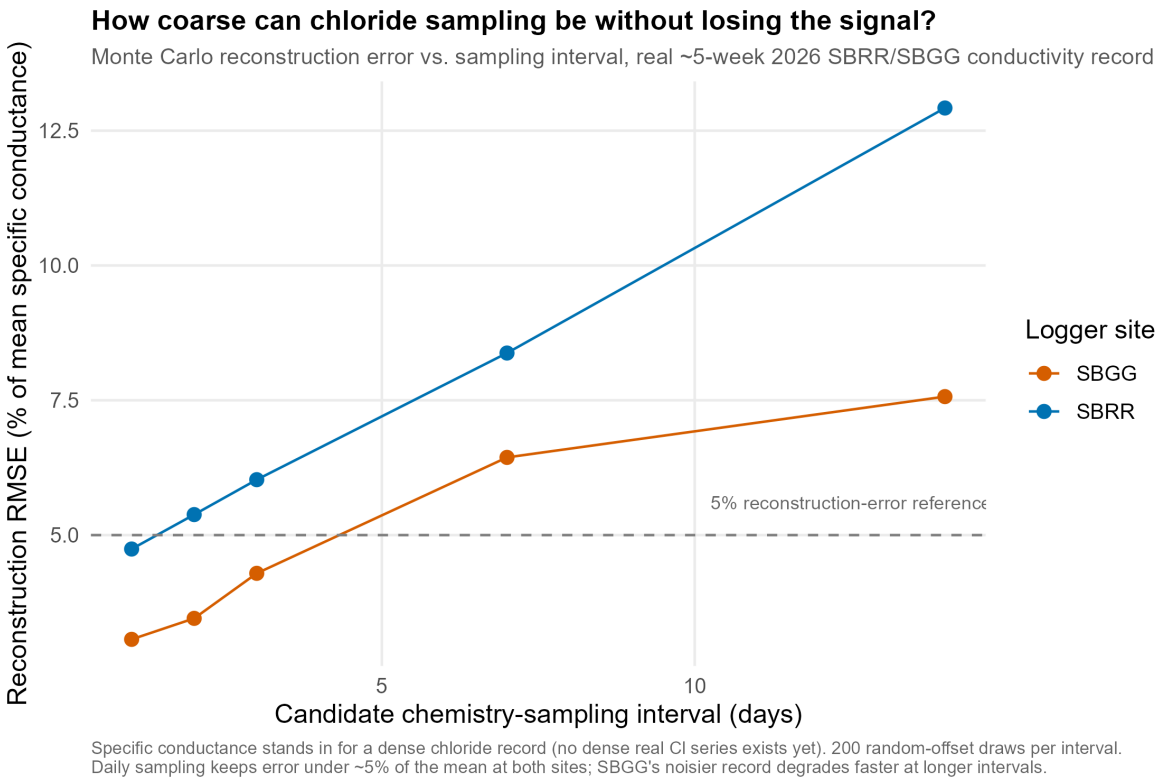


Figure 5: Monte Carlo reconstruction-error curve: average error (with spread across 200 randomized trials per interval) vs. candidate chemistry sampling interval, with a 5% reference line.

Honest caveat, stated plainly: this whole pipeline is built, tested, and working end-to-end – but it has only ever been run on synthetic (made-up, but realistic) chloride data so far, because the real chloride lab samples collected to date (2026-05-01) predate the conductivity logger deployment (2026-07-15) by about 2.5 months, so there's no real overlap yet. The method is proven; it just needs real, time-overlapping chloride samples to produce a real recommendation instead of a demonstration. That is the direct motivation for the calibration sampling campaign proposed in the thesis (monthly chloride sampling for a full year, starting as soon as possible).

Questions for you

- Does the pH gap (injection vs. shallow network) support framing this as reactive transport rather than conservative mixing, at least for near-surface processes?
- How would you interpret two wells (Herz Deep, NDOT) trending in opposite, uncorrelated directions on temperature vs. conductivity over the same window?

- Does the Monte Carlo sampling-frequency design above seem like the right level of rigor for a thesis-scale calibration campaign, or would you want it extended/simplified before the real field season starts?
- Given real water-level coverage *inside* the Steamboat field itself is currently only 2-3 wells, what's the right first modeling step – a simple mixing/box model, or is it worth waiting for better spatial coverage?